



Micro:bit Blocks

My name _____

Class _____

Date _____



How do we give a tiny computer instructions?

A micro:bit is a tiny computer. It does exactly what your code tells it. Let's give it instructions with blocks!



My kit tick when you have it

- ☐ A micro:bit and a USB cable
- ☐ MakeCode open on a tablet or laptop
- ☐ MakeCode



Stay safe

- Hold the micro:bit by its edges.
- Keep the micro:bit and battery pack dry.
- An adult plugs in the cable.

START HERE: BUILD



Make a smiley face

1 · DO IT: TICK EACH STEP

- 1 ☐ Find the "on start" block.
- 2 ☐ Drag "show icon" inside "on start".
- 3 ☐ Pick the smiley face.
- 4 ☐ Ask an adult to download it to the micro:bit.
- 5 ☐ Look at the lights on the micro:bit.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	What does the micro:bit show?
My micro:bit	A smiley face ? Something else Nothing



Draw what the lights show

WORDS TO USE

code block program lights smiley download

TEST 1

1234

Button counter

? If you press A 10 times, what number will it show?

1 · MY GUESS: CIRCLE ONE

10

Exactly 10



More than 10



Less than 10

How sure am I? 🤔 😊 😎

2 · DO IT: TICK EACH STEP

- 1 ☐ 📦 Make a variable called "count".
- 2 ☐ 📌 A Use "on button A pressed".
- 3 ☐ ➕ Inside it, put "change count by 1".
- 4 ☐ 📄 12 34 Then put "show number count".
- 5 ☐ 🖱️ Download it. Press A slowly, 10 times.

3 · LOOK & RECORD

	📄 12 34 What number does it show?
📌 A After 10 presses	

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right ☐ 🌑 Partly right ☐ 🔄 Something different happened ☐ 🤔 Not sure yet

WORDS TO USE

button variable count press number correct bug

TEST 2 🔧 Make a thermometer

? Will the micro:bit be warmer on your desk or in your hands?

1 · MY GUESS: CIRCLE ONE

- ☐ 🪑 On the desk ☐ 👐 In my hands ☐ ⚖️ Both the same

How sure am I? 🤔 😊 😎

2 · DO IT: TICK EACH STEP

- 1 ☐ 📌 B Use "on button B pressed".
- 2 ☐ 🌡️ Inside it, put "show number temperature".
- 3 ☐ 🖱️ Download it. Put the micro:bit on your desk.
- 4 ☐ ⌚ Wait 2 minutes. Press B and read it.
- 5 ☐ 👐 Hold it gently in your hands for 2 minutes.
- 6 ☐ 📌 B Press B and read it again.

3 · LOOK & RECORD

Test	🌡️ What does it show?
🪑 On the desk	°C
👐 In my hands	°C

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right ☐ 🌑 Partly right ☐ 🔄 Something different happened ☐ 🤔 Not sure yet

WORDS TO USE

temperature sensor warm cool hands degrees

Art corner: Pixel picture

Colour squares in the 5×5 grid to design your own picture. Then build it with the "show leds" block.

Maths corner: Counting code

Press A 3 times. What number should it show?
Draw it.

New words

Program: Instructions that tell a computer what to do.

Bug: A mistake in code.

Code: The instructions we write for a computer.

Sensor: A part that notices something, like heat or light.

Variable: A box in a program that remembers something.

What I can do now colour the stars

I can put blocks in order to make a program.



I can find and fix a bug.



I can use a sensor to measure something.

